Trade Compliance Copilot

AI-Powered Trade Surveillance

Demo: Intelligent Detection, Explanation & Human Review

Detection Pipeline

Simulate Trades: Generate realistic trading activity to test and validate detection algorithms in a controlled environment.

Rule-Based Detection: Apply predefined rules to identify suspicious patterns like wash trades and spoofing behaviors in real-time.

Embedding Similarity Boost: Enhance detection accuracy by comparing trade patterns against known violations using vector similarity in LanceDB.

Confidence & Routing

Each detected anomaly receives a confidence score (0-100%) based on rule severity and embedding similarity. High-confidence alerts (>80%) route directly to senior compliance officers, while medium scores (50-80%) go to standard review queues. Low-confidence detections are logged for pattern analysis but don't trigger immediate action.

LLM Explanation Engine

Once a trade is flagged, the LLM generates clear, human-readable explanations for each alert.
The engine analyzes pattern context, rule triggers, and similarity scores to produce detailed narratives that help compliance officers quickly understand why a trade was flagged and assess its risk level.

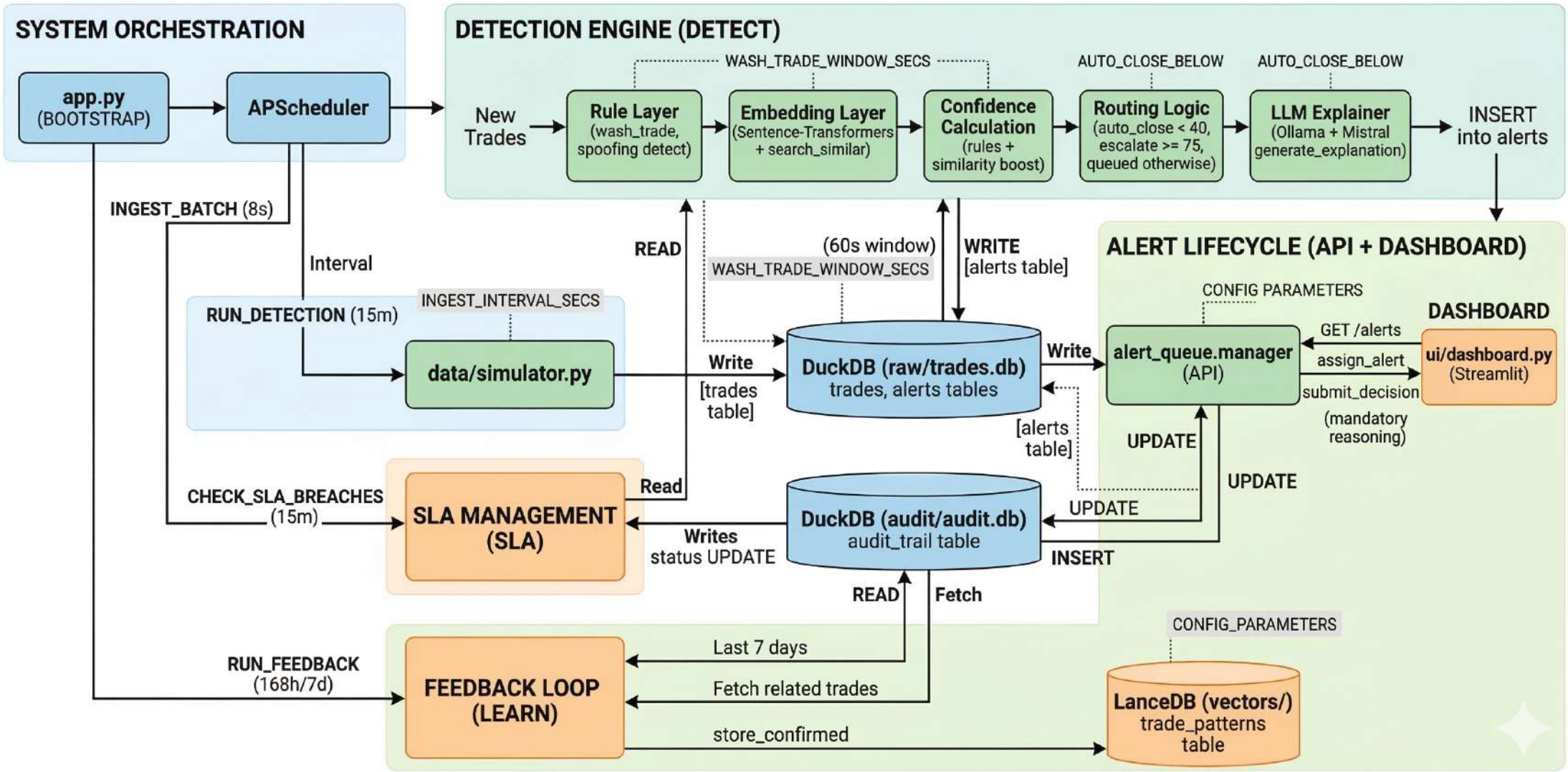
Review & Audit

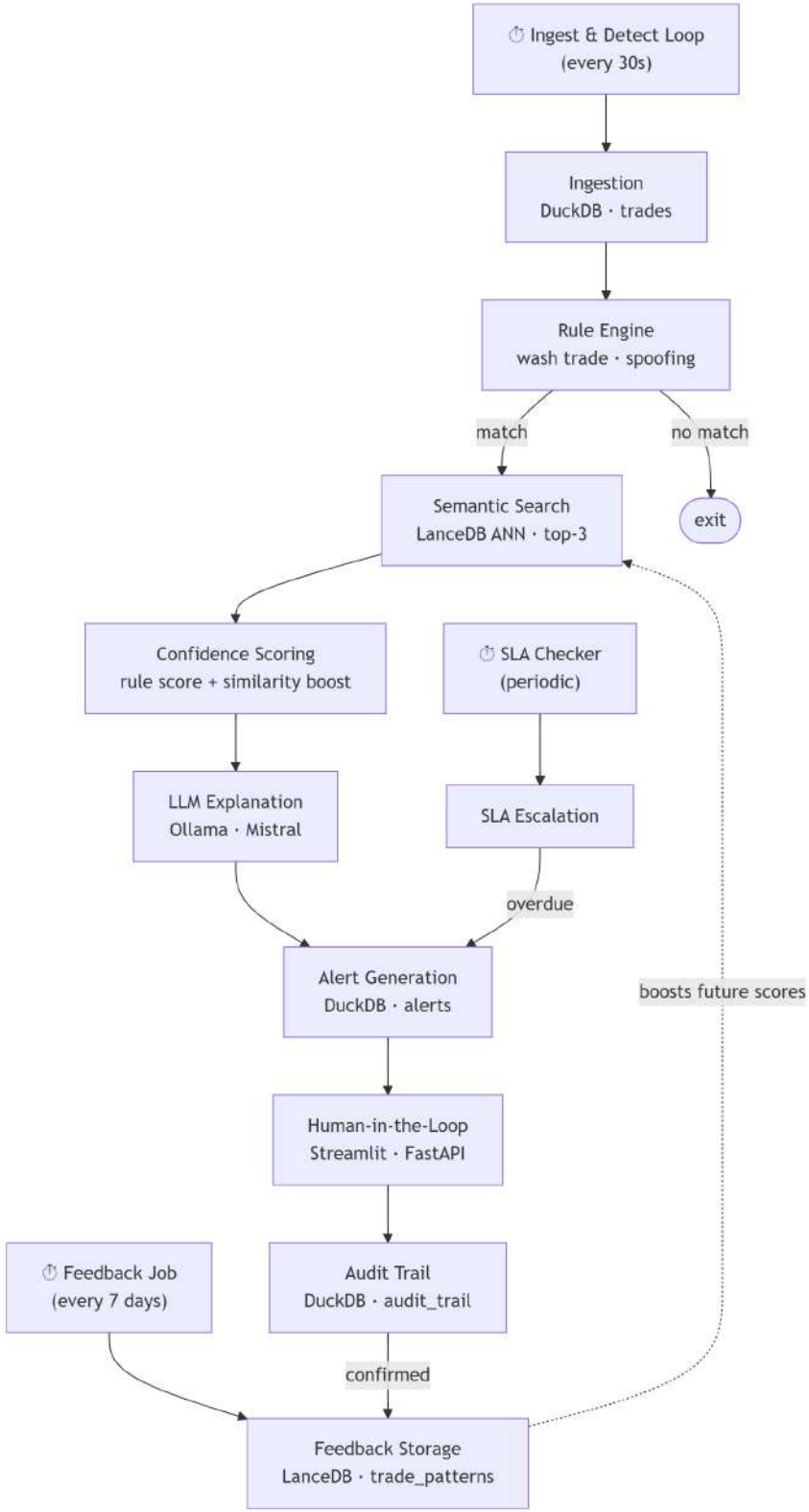
Persist alerts to a centralized database for tracking, investigation, and regulatory reporting requirements. Human compliance officers review flagged trades, validate AI-generated explanations, and make final disposition decisions. Comprehensive audit logging captures every action, decision, and timestamp to ensure full regulatory accountability.

Continuous Learning Loop

The system grows smarter over time through periodic feedback integration. Human review decisions are embedded and stored in the LanceDB vector library, continuously improving future similarity detection accuracy and reducing false positives.

'TRADE COMPLIANCE COPILOT: HIGH-LEVEL ARCHITECTURE & DATA FLOW




⏱ Ingest & Detect Loop
(every 30s)
Ingestion
DuckDB · trades
Rule Engine
wash trade · spoofing
match
no match
exit
Semantic Search
LanceDB ANN · top-3
Confidence Scoring
rule score + similarity boost
⏱ SLA Checker
(periodic)
LLM Explanation
Ollama · Mistral
SLA Escalation
overdue
Alert Generation
DuckDB · alerts
boosts future scores
Human-in-the-Loop
Streamlit · FastAPI
⏱ Feedback Job
(every 7 days)
Audit Trail
DuckDB · audit_trail
confirmed
Feedback Storage
LanceDB · trade_patterns

Deploy

Compliance Copilot

Officer ID

officer-01

View

- Alert Queue
- Review Alert
- Audit Trail
- Statistics
- Admin

Open Closed

Alert Queue

ALT-E3D1C284

TRD-003 · AMZN

WASH TRADE

UNDER_REVIEW

55

Review

Deploy

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View

- Alert Queue
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Open 5

Closed 0

Select Alert

ALT-E3D1C284 · TRD-003 · AMZN

ALT-E3D1C284 · TRD-003 · AMZN

ALT-9BAA5897 · TRD-001 · TSLA

ALT-D402375F · TRD-001 · GS

ALT-CCF31994 · TRD-001 · GS

ALT-299DBCF3 · TRD-001 · TSLA

Or enter Alert ID

AI Analysis

The compliance officer should investigate trade activity by TRD-003 on AMZN for potential wash trading. There is evidence of a single buy-sell pair within a 30-second interval, with an average price difference of 0.0042. The confidence level is 55%, and there are no similar historical cases to reference.

Deploy

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Open 5

Closed 0

The compliance officer should investigate trade activity by TRD-003 on AMZN for potential wash trading. There is evidence of a single buy-sell pair within a 30-second interval, with an average price difference of 0.0042. The confidence level is 55%, and there are no similar historical cases to reference.

Trader History

alert_id	instrument	pattern	confidence	status	decision
ALT-E3D1C284	AMZN	wash_trade	55	under_review	None

Submit Decision

Decision is permanent and recorded in the audit trail.

Decision

✗ Confirmed — Real Manipulation

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✓ False Positive — Clear Alert

↑ Escalate — Needs Senior Review

Compliance Copilot

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Open Closed

Audit Trail

Immutable record of all compliance decisions.

alert_id	trader_id	pattern	confidence_at_decision	decision	officer_id	time_to_decision_secs	recorded_at
ALT-3ECE1C85	TRD-003	wash_trade	55	confirmed	officer-01	6s	2026-03-13 21:43

Compliance Copilot

Officer ID

officer-01

View

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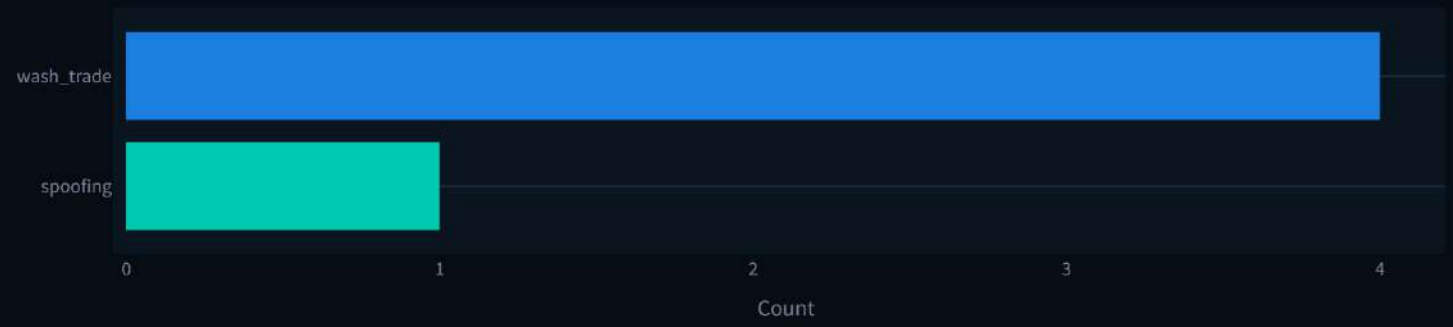
Statistics

Total Alerts	Open	Closed
5	3	2

Alerts by Pattern